# Yuanpeng Li

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**Research:** Reinforcement learning (RL) and multi-task optimization, from small control policies to VLA models and LLMs

## EDUCATION

**University of California, Irvine** — **Irvine, CA**
*Doctor of Philosophy (PhD) in Statistics; Advisors: Prof. Annie Qu, Prof. Rui Miao* — *Sep. 2024 – May 2028 (expected)*

**Jilin University – School of Mathematics** — **Changchun, China**
*B.S. in Mathematics and Applied Mathematics* — *Aug. 2020 – Jul. 2024*

## PUBLICATIONS

*TOPPO: Rethinking PPO for Multi-Task Reinforcement Learning with Critic Balancing.* — *NeurIPS 2026*
**Yuanpeng Li**, Gefei Lin, Annie Qu, Rui Miao. arXiv:2605.11473.

## RESEARCH EXPERIENCE

**TOPPO: Rethinking PPO for Multi-Task RL with Critic Balancing** [code] — **Irvine, CA**
*First author; accepted at NeurIPS 2026; Advisors: Prof. Annie Qu, Prof. Rui Miao* — *Apr. 2026*

- Identified ill-conditioned critic gradients as the key bottleneck in multi-task PPO: easy tasks dominate the critic and hard tasks stall.
- Proposed Critic Balancing (per-task PopArt, LayerNorm critic, FairGrad on the critic, PCGrad on the actor), proved that FairGrad's $\alpha{=}1$ update is unchanged by positive per-task gradient rescaling, and developed a projected Newton solver 83× faster than SciPy.
- Set a new state of the art on Meta-World+ MT50 (90.9% mean, 56.5% worst-10-task success) over the strongest baseline, ARS-LN-1024 (85.7%, 43.3%), with 22.7× fewer parameters, reaching the baseline's final mean in 28% of the steps.
- Extended Critic Balancing to a two-objective AMP humanoid task, cutting motion-style FID by 76%.

**Scaling TOPPO to Large Models: VLA and LLM RL** — **Irvine, CA**
*Advisors: Prof. Annie Qu, Prof. Rui Miao* — *Sep. 2026*

- Set up RL fine-tuning of a 3.3B $\pi_0$ VLA model with RLinf, and built a GRPO/PPO pipeline for LLMs on verl, vLLM and FSDP2, with a Nemotron-3-Super-120B judge self-hosted on vLLM as the reward model.
- Standard whole-network gradient surgery gave no clear gain on $\pi_0$ or LLMs, which led to the block-wise study below.

**Block-Wise Gradient Balancing for Deep Multi-Task Networks** — **Irvine, CA**
*Advisors: Prof. Annie Qu, Prof. Rui Miao* — *Ongoing*

- Benchmarked 26 multi-task gradient methods in one PyTorch trainer, reproducing published NYUv2, CelebA and QM9 results.
- Found that a few small early layers carry most of the gradient, so the whole-network Gram matrix FairGrad uses to weight tasks mostly reflects them; yet on QM9 the energy targets' gradients differ only in the final readout layer, which this matrix barely sees.
- Proposed a norm-induced block kernel that balances each module's contribution to FairGrad's task weights. On a QM9 graph transformer it cut $\Delta m\%$ (average gap to single-task models) from 27.7 to 9.6.

**Runtime-Aware HPC Job Scheduling** — **Irvine, CA**
*With Prof. Rui Miao* — *Ongoing*

- Wrote slurmsim, a Python/C Slurm simulator matching real Slurm's job start order and 99.9% of sampled priorities exactly.
- Set each job's `--time-min`, which backfill plans with, from a Kaplan–Meier runtime estimate, keeping the user's limit as the hard cap; mean turnaround fell 8–18% in simulation and 8% in a national-lab trace replay, with almost no extra timeouts.

## PROJECTS

**RunFleet** & **Run-Skills** | *Python, Bash, Slurm, MCP*

- Built a terminal dashboard for jobs and GPUs across Slurm clusters over SSH, plus skills that let coding agents submit jobs.

**Gradescope MCP** | *Python, MCP, uv*

- Built an MCP server that lets AI agents grade and edit Gradescope rubrics; every write is previewed and approved by a person.

## TEACHING

- **Graduate Teaching Assistant**, UC Irvine Statistics (Sep. 2024 – Present): STATS 7, 67, 110, 120C, 205P.

## AWARDS

- UC Irvine Statistics **Graduate Travel Award**, $2,500 (2026); Chinese Mathematics Competitions **National 3rd Prize** (2021)

## SKILLS

- **Methods:** Reinforcement learning (PPO, GRPO), RLHF/RLAIF, multi-task learning (PCGrad, FairGrad, MGDA), LoRA/PEFT
- **ML Frameworks & Tooling:** PyTorch, Hugging Face, vLLM, verl, FSDP, RLinf; MCP servers, Claude Code / Codex skills
- **Infra & Engineering:** Python, C, Git, Docker, Linux, Slurm, Ray, W&B, uv, pytest